PERSONAL PLANNING AGENT

Member 3: Schedule Optimization & Task Execution Engine

AGENT 3 OF 5

1. AGENT OVERVIEW & CORE PURPOSE

The **Personal Planning Agent** translates abstract academic and personal goals into optimized, actionable daily schedules. Operating with functional tool-use capabilities, it handles schedule constraints, task prioritization, and dynamic conflict resolution without manual user intervention.

Primary Objective

Build and maintain a dynamic schedule that balances academic requirements, life commitments, and user fatigue constraints.

Tool-Driven Execution

Executes programmatic tools (e.g., `create_task()`, `reschedule_block()`) to directly modify the user's operational calendar.

2. UNIVERSAL SYSTEM DATA SCHEMA CONTEXT

The Planning Agent owns the schedule structure inside `study_data` and reads constraint parameters from `adaptive_model`:


```
{ "study_data": { "university_schedule": [ { "day": "Monday", "start": "09:00", "end": "12:00", "title": "CS101 Lecture" } ], "tasks": [ { "task_id": "T101", "title": "Trees Practice", "estimated_mins": 45, "status": "pending" } ] }, "adaptive_model": { "peak_productivity_window": "09:00-11:00", "avg_task_completion_ratio": 0.75 } }
```


3. INPUT & OUTPUT SPECIFICATION

Type	Input Sources / Formats	Outputs / Artifacts Produced
Data Payload	• University timetables & hard deadlines • Goal milestones from **Life Coach Agent** • User energy/productivity profile	• Conflict-free daily/weekly timetables • Executed schedule updates & task allocations • Workload density metrics
Control Signals	• Reschedule directives from **Adaptive Manager** due to observed task avoidance or delays	• Schedule feasibility flags sent to Adaptive Manager if deadlines become unreachable

4. INTERNAL EXECUTION & OPTIMIZATION FLOW

The Planning Agent evaluates constraints through a multi-step execution pipeline:

1. **Constraint Ingestion:** Reads fixed calendar commitments (lectures, work) and high-priority deadlines.
2. **Task Deconstruction & Prioritization:** Breaks large milestones into micro-tasks matching the user's attention span.
3. **Schedule Optimization:** Fits tasks into high-energy cognitive windows based on peak productivity metrics.
4. **Tool Execution:** Calls internal tools (`create_task()`, `optimize_schedule()`) to lock calendar blocks.

5. INTER-AGENT WORKFLOW & INTEGRATION ARCHITECTURE

The Planning Agent converts higher-level decisions from other agents into time-blocked reality:

Inbound Interactions

- **Goal & Life Coach Agent:** Provides high-level goal decompositions requiring schedule slots.
- **Adaptive Manager:** Sends preemptive restructuring commands to avoid anticipated burnout.

Outbound Interactions

- **AI Tutor Agent:** Triggers scheduled study sessions at designated block start times.
- **Adaptive Manager:** Alerts manager when user misses schedule blocks repeatedly.

6. DETAILED AGENT COMMUNICATION MATRIX

Target Agent	Data Shared / Received	Trigger Condition / Purpose
Adaptive Manager	Completion Rates & Schedule Slip Metrics	Informs manager when real completion velocity deviates from plan by > 20%.
Goal & Life Coach Agent	Milestone Feasibility Reports	Alerts Coach if user calendar lacks capacity for ambitious career goals.
AI Tutor Agent	Study Block Invitations & Context Payload	Pushes scheduled study session triggers along with target topic references.

7. VALUE PROPOSITION & SYSTEM IMPACT

The Personal Planning Agent bridges the gap between intention and execution:

- **Dynamic Real-Time Adaptation:** Replaces rigid calendar systems with a flexible schedule that adapts seamlessly to actual execution velocity.
- **Proactive Fatigue Management:** Automatically prevents task overcrowding by respecting productivity windows in the user's personal profile.
- **Actionable Tool Execution:** Executes actual calendar modifications directly, eliminating manual user organization overhead.